

A TEST WHETHER TWO AIC'S DIFFER SIGNIFICANTLY

H. Linhart : Institut für Statistik und Ökonometrie, Universität
Göttingen, Platz der Göttinger Sieben 5, D-3400
Göttingen, F.R. Germany.

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Summary : An approximate test of the hypothesis that one model
leads to a smaller expected (Kullback-Leibler) discrepancy
than another one is derived. It is based on the asymptotic
distribution of the difference between the two empirical
discrepancies under misspecification for both models.

1. Introduction

Akaike's information criterion (AIC) is an estimator of the expected Kullback-Leibler discrepancy between the operating model and a fitted (parametric) approximating model. It is used for the selection of one out of two (or more) candidate models: the model leading to the smaller criterion is preferred.

Model selection - on the basis of expected discrepancies - can either be carried out by *estimating* the expected discrepancies of the two candidates or by *testing* the hypothesis that one candidate has a larger expected discrepancy than the other. In this note we give a test for the case in which

the Kullback-Leibler discrepancy is the basis of the analysis.

2. Notation and regularity conditions

Let $F(x)$ and $G_\theta(x)$, $\theta \in \Theta \subset R^p$, $x \in R^k$, be the operating and approximating distribution function. The Kullback-Leibler discrepancy is then

$$\Delta(\theta) = -E_F \log g_\theta(x),$$

where g_θ is the density of G_θ .

We use the notation

$$\theta_0 = \arg \min_{\theta \in \Theta} \Delta(\theta)$$

$$\Delta_n(\theta) = -E_{F_n} \log g_\theta(x) = -\frac{1}{n} \sum_{\nu=1}^n \log g_\theta(x_\nu)$$

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \Delta_n(\theta).$$

Here $\{x_\nu : \nu = 1, 2, \dots, n\}$ constitutes a random sample from F , F_n is the empirical distribution function and $\hat{\theta}_n$ is the maximum likelihood estimator.

We assume the following regularity conditions (White (1982), Linhart and Zucchini (1986)):

G_θ has a density $g_\theta(x)$ for all $\theta \in \Theta$, $x \in R^k$. $\log g_\theta(x)$ is twice continuously differentiable in θ for all x ; $\log g_\theta(x)$ and its two derivatives

are measurable functions of x for all θ and their moduli are denominated by functions of x only which are integrable w.r.t. F . Θ is convex and compact, θ_0 is unique and in the interior of Θ , $\Delta''(\theta_0)$ is positive definite.

Under these conditions follows that $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \Omega^{-1}\Sigma\Omega^{-1})$, where

$$\Omega = \Delta''(\theta_0) = -\{E_F \partial^2 \log g_{\theta_0}(x) / \partial \theta_r \partial \theta_s : r, s = 1, 2, \dots, p\}$$

and

$$\Sigma = \{E_F (\partial \log g_{\theta_0}(x) / \partial \theta_r) (\partial \log g_{\theta_0}(x) / \partial \theta_s) : r, s = 1, 2, \dots, p\},$$

and also that

$$\Delta_n(\hat{\theta}_n) + \text{tr } \Omega_n^{-1} \Sigma_n / n$$

and

$$\Delta_n(\hat{\theta}_n) + p/n$$

are reasonable estimators of $E_F \Delta(\hat{\theta}_n)$ (Akaike (1973), White (1982), Linhart and Zucchini (1986)).

3. The test

Assume now that two families of models $G_{\theta}^{(i)}$, $i = 1, 2$, are candidates for selection. We derive - under the regularity conditions and under misspecification for both models - an approximate test of the hypothesis that

$$E\Delta^{(1)}(\hat{\theta}_n^{(1)}) \leq E\Delta^{(2)}(\hat{\theta}_n^{(2)}).$$

For this we need at first the joint asymptotic distribution of

$$\sqrt{n}(\Delta_n^{(i)}(\hat{\theta}_n^{(i)}) - \Delta^{(i)}(\theta_0^{(i)})), \quad i = 1, 2.$$

It is sufficient to consider $\sqrt{n}(\Delta_n(\theta_0) - \Delta(\theta_0))$ because $\sqrt{n}(\Delta_n(\hat{\theta}_n) - \Delta_n(\theta_0)) = \sqrt{n}(\hat{\theta}_n - \theta_0)\Delta'_n(\bar{\theta}_n) = o_p(1)$. Here $|\bar{\theta}_n - \theta_0| \leq |\hat{\theta}_n - \theta_0|$ (Jennrich (1969), Lemma 3), $\sqrt{n}(\hat{\theta}_n - \theta_0) = o_p(1)$, $\Delta'_n(\theta) \xrightarrow{\text{a.s.}} \Delta'(\theta)$ uniformly in θ (Jennrich (1969), Theorem 2), thus $\Delta'_n(\hat{\theta}_n) \xrightarrow{\text{a.s.}} \Delta'(\theta_0) = 0$.

By the central limit theorem

$$\begin{pmatrix} \sqrt{n}(\Delta_n^{(1)}(\theta_0^{(1)}) - \Delta^{(1)}(\theta_0^{(1)})) \\ \sqrt{n}(\Delta_n^{(2)}(\theta_0^{(2)}) - \Delta^{(2)}(\theta_0^{(2)})) \end{pmatrix} \xrightarrow{d} N(0, \Lambda)$$

where $\Lambda = \{\lambda_{ij} : i, j = 1, 2\}$ and

$$\lambda_{ij} = E_F \log g_{\theta_0^{(i)}}^{(i)}(x) \log g_{\theta_0^{(j)}}^{(j)}(x) - E_F \log g_{\theta_0^{(i)}}^{(i)}(x) E_F \log g_{\theta_0^{(j)}}^{(j)}(x).$$

Thus

$$\frac{\Delta_n^{(1)}(\hat{\theta}_n^{(1)}) - \Delta_n^{(2)}(\hat{\theta}_n^{(2)}) + (\Delta^{(1)}(\theta_0^{(1)}) - \Delta^{(2)}(\theta_0^{(2)}))}{((\lambda_{11} + \lambda_{22} - 2\lambda_{12})/n)^{1/2}} \xrightarrow{d} N(0, 1).$$

By proposition 1 of Linhart and Zucchini (1986, section 2.4) the expected discrepancies are approximately equal to $\Delta^{(i)}(\theta_0^{(i)}) + \frac{1}{2n} \text{tr } \Omega^{(i)-1} \Sigma^{(i)}$ or, if one wants to approximate $\text{tr } \Omega^{-1} \Sigma$ by

$p = \dim(\theta)$, to $\Delta^{(i)}(\theta_0^{(i)}) + \frac{p^{(i)}}{2n}$.

Under the hypothesis that the two expected discrepancies are equal

$$\frac{\Delta_n^{(1)}(\hat{\theta}_n^{(1)}) - \Delta_n^{(2)}(\hat{\theta}_n^{(2)}) + p^{(1)}/2n - p^{(2)}/2n}{((\lambda_{11} + \lambda_{22} - 2\lambda_{12})/n)^{1/2}} \xrightarrow{d} N(0, 1).$$

The same asymptotic distribution holds if $p^{(i)}/2n$ is replaced by $p^{(i)}/n$ and one has that - under the mentioned hypothesis -

$$\frac{AIC(1) - AIC(2)}{((\lambda_{11} + \lambda_{22} - 2\lambda_{12})/n)^{1/2}} \xrightarrow{d} N(0, 1).$$

Consistent estimators $\hat{\lambda}_{ij}$ of λ_{ij} are obtained if in the expressions for λ_{ij} the parameters $\theta_0^{(i)}$ are replaced by $\hat{\theta}_0^{(i)}$ and F by the empirical d.f. F_n . The asymptotic distribution remains unchanged if λ_{ij} is replaced by

$$\hat{\lambda}_{ij} = \frac{1}{n} \sum_{\nu=1}^n \log g_{\hat{\theta}_n^{(i)}}^{(i)}(x_\nu) \log g_{\hat{\theta}_n^{(j)}}^{(j)}(x_\nu) - \Delta_n^{(i)}(\hat{\theta}_n^{(i)}) \Delta_n^{(j)}(\hat{\theta}_n^{(j)})$$

because $\hat{\lambda}_{ij} \xrightarrow{a.s.} \lambda_{ij}$.

The approximate test is carried out by the $N(0,1)$ test statistic

$$z = n^{1/2}(AIC(1) - AIC(2))/(\hat{\lambda}_{11} + \hat{\lambda}_{22} - 2\hat{\lambda}_{12})^{1/2},$$

one-tailed or two-tailed. The two-tailed test could also use z^2 which can be treated as chi-squared with one degree of freedom.

4. Remarks

4.1

The results of the last section could also be used to give approximate confidence intervals for expected Kullback-Leibler discrepancies. If z_α is the $100(1-\alpha)$ per cent point of $N(0,1)$ an approximate confidence interval for the expected discrepancy under model i is

$$(\text{AIC}(i) - z_\alpha(\hat{\lambda}_{ii}/n)^{1/2}, \text{AIC}(i) + z_\alpha(\hat{\lambda}_{ii}/n)^{1/2}).$$

The corresponding interval for the difference between two expected discrepancies is

$$(D - z_\alpha s, D + z_\alpha s),$$

where $D = \text{AIC}(1) - \text{AIC}(2)$ and $s^2 = (\hat{\lambda}_{11} + \hat{\lambda}_{22} - 2\hat{\lambda}_{12})/n$.

4.2

If the two models are *nested* an alternative test could be carried out using results of Steiger, Shapiro and Browne (1985). Their theorem 1 states that the asymptotic distribution of $n(\Delta_n^{(1)}(\hat{\theta}_n^{(1)}) - \Delta_n^{(2)}(\hat{\theta}_n^{(2)}))$ is noncentral chi-squared with noncentrality parameter $\Delta^{(1)}(\theta_0^{(1)}) - \Delta^{(2)}(\theta_0^{(2)})$. This could be used to test hypotheses (or to construct confidence intervals) about the noncentrality parameter and hence also about the difference in expected discrepancies.

The results of Steiger *et al.* are obtained under "local alternatives", sequences of operating models are used which converge to one of the approximating models. Our results are derived for "fixed alternatives".

This explains why their results are not special cases (for nested models) of ours. To us fixed alternatives seem to be the more appropriate set up than local alternatives but only Monte Carlo experiments could show which asymptotic approach yields better approximations to the finite sample case.

5. Examples of applications

5.1

Linhart and Zucchini (1986, p. 59-61) fit a distribution to the annual flow of the Vaal river at Standerton. They attempt the normal, lognormal and gamma distributions.

The values of Akaike's information criterion are:

Normal	$7,516 + 2/65 = 7,547$
Lognormal	$7,153 + 2/65 = 7,184$
Gamma	$7,191 + 2/65 = 7,222$

The normal distribution has many advantages and one could be of the opinion that it should be chosen unless there is strong evidence that one of the other distributions is preferable.

One tests the hypothesis that the normal distribution leads to a smaller expected discrepancy than the lognormal distribution. This is a one-sided test. One obtains $\hat{\lambda}_{11} = 1,8765$, $\hat{\lambda}_{12} = 1,0516$, $\hat{\lambda}_{22} = 0,9854$ and calculates

$$z = (7,547 - 7,184)(65)^{1/2} / (1,8765 + 0,9854 - 2 \cdot 1,0516)^{1/2} = 3,36.$$

Such a value or a larger one is very rare under $N(0,1)$, the hypothesis must be rejected. The normal distribution must not be used.

The difference between the lognormal and the gamma distribution is not significant. One tests two-sided the hypothesis that the two expected discrepancies are equal by means of

$$\chi^2 = (7,222 - 7,184)^2 65 / (1,0160 + 0,9854 - 2 \cdot 0,9740) = 1,75.$$

Under χ^2_1 this result is not significant.

5.2

In the mentioned book (p. 200-203) a contingency table is considered which resulted from a survey on the opinion of inhabitants of Göttingen on a proposed refuse collection scheme. Various loglinear models were judged by their AICs and model 2 with sufficient configurations (ij) and (jk) was selected. An interesting alternative would be model 7 with sufficient configurations (i) and (jk) , which has a $G^2 + 2p$ of 62,18 (as compared with 55,14 for model 2). Model 7 is less complex and one could be of the opinion that it should be preferred to model 2 unless the hypothesis that model 2 is worse can be rejected at a small significance level.

For contingency tables one frequently uses $G^2 + 2p$ as "AIC". This form is equivalent for model selection purposes, but if the given test is applied one must note that $G^2 + 2p = 2n(\text{AIC} + \sum_{ijk} (n_{ijk}/n) \log(n_{ijk}/n))$. If one wants to use $G^2 + 2p$ the test statistics in section 3 must be modified: $\text{AIC}(1) - \text{AIC}(2)$ must be replaced by $(1/2n)[(G^2(1) + 2p^{(1)}) - (G^2(2) + 2p^{(2)})]$.

For the comparison of model 7 with model 2 one gets $\hat{\lambda}_{11} = 0,9431$, $\hat{\lambda}_{12} = 0,9337$, $\hat{\lambda}_{22} = 0,9345$ and $z = 1,00$ which has a P -value of 0,16 in a one-tailed test. One would use model 7.

Of all other models only the one with sufficient configurations (i) , (j) and (k) seems a possibility. But the hypothesis that this is a better model than 7 must be rejected ($\hat{\lambda}_{11} = 0,9100$, $\hat{\lambda}_{12} = 0,8941$, $\hat{\lambda}_{22} = 0,9431$, $z = 4,42$).

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